

## Тема 3. Управление БД в веб-приложении

Для создания базы данных MongoDB можно использовать MongoDB Cloud.

Подключение к серверу баз данных может производиться с помощью утилиты mongosh (MongoDB Shell).

### 1. Создание базы данных

```
use j_users
```

### 2. Добавление содержимого

```
db.users.insertMany([{"login":"student","password":"tnecluts"}, {"login":"myuser","password":"mypas"}, {"login":"teacher","password":"qq"}, {"login":"myking","password":"myqueen"}])
```

### 1. Получение данных

```
db.users.findOne({login: "teacher"}).password
```

### 1. Подключение с помощью mongosh

```
mongosh "mongodb connection string"
```

### 1. Получение списка всех записей

```
db.users.find().map(x=>({username:x.login, password:x.password}))
```

```
Atlas atlas-jastie-shard-0 [primary] j_users> db.users.find().map(x => ({ username: x.login, password: x.password }))
[
  { username: 'student', password: 'tnecluts' },
  { username: 'myuser', password: 'mypas' },
  { username: 'teacher', password: 'qq' },
  { username: 'myking', password: 'myqueen' }
]
Atlas atlas-jastie-shard-0 [primary] j_users> |
```

The screenshot displays the MongoDB Atlas web interface. The top navigation bar includes the Atlas logo, a dropdown menu for 'Шульга's Org...', and links for 'Access Manager', 'Billing', 'All Clusters', 'Get Help', and a user profile 'Шульга'. The main sidebar on the left lists various services under 'Project 0', including 'Database', 'Data Lake', 'SERVICES', 'Device & Edge Sync', 'Triggers', 'Data API', 'Data Federation', 'Atlas Search', 'Stream Processing', 'Migration', 'SECURITY', 'Quickstart', 'Backup', 'Database Access' (highlighted), 'Network Access', 'Advanced', and 'New On Atlas'. The main content area is titled 'Database Access' and shows a table of 'Database Users'. The table has columns for 'User Name', 'Authentication Method', 'MongoDB Roles', 'Resources', and 'Actions'. Two users are listed: 'eugeneshulga03' and 'student', both using 'SCRAM' authentication and having the role 'readWriteAnyDatabase@admin'. The 'Resources' column shows 'All Resources' for both. The 'Actions' column contains 'EDIT' and 'DELETE' buttons for each user. A green button '+ ADD NEW DATABASE USER' is located at the top right of the table. At the bottom of the page, the system status is 'All Good', and the footer includes copyright information for MongoDB, Inc. and links to 'Status', 'Terms', 'Privacy', 'Atlas Blog', and 'Contact Sales'.

| User Name      | Authentication Method | MongoDB Roles              | Resources     | Actions                                     |
|----------------|-----------------------|----------------------------|---------------|---------------------------------------------|
| eugeneshulga03 | SCRAM                 | readWriteAnyDatabase@admin | All Resources | <a href="#">EDIT</a> <a href="#">DELETE</a> |
| student        | SCRAM                 | readWriteAnyDatabase@admin | All Resources | <a href="#">EDIT</a> <a href="#">DELETE</a> |

Project 0

Data Services

App Services

Charts

Data Lake

SERVICES

Device & Edge Sync

Triggers

Data API

Data Federation

Atlas Search

Stream Processing

Migration

SECURITY

Quickstart

Backup

Database Access

Network Access

Advanced

New On Atlas 6

Goto

We are deploying your changes (current action: creating a plan)

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Network Access

IP Access List

Peering

Private Endpoint

+ ADD IP ADDRESS

You will only be able to connect to your cluster from the following list of IP Addresses:

| IP Address                                         | Comment                                   | Status  | Actions                           |
|----------------------------------------------------|-------------------------------------------|---------|-----------------------------------|
| 0.0.0.0/0 (includes your current IP address)       |                                           | Pending | <div>EDIT</div> <div>DELETE</div> |
| 23.104.56.11/32 (includes your current IP address) | Created as part of the Auto Setup process | Active  | <div>EDIT</div> <div>DELETE</div> |

System Status: All Good

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```
mongosh mongodb+srv://<cr>
Connecting to: mongodb+srv://<credentials>@cluster0.cflipbf.mongodb.net/?appName=mongosh+2.0.0
Using MongoDB: 7.0.11 (API Version 1)
Using Mongosh: 2.0.0
mongosh 2.2.10 is available for download: https://www.mongodb.com/try/download/shell
For mongosh info see: https://docs.mongodb.com/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

Atlas atlas-jastie-shard-0 [primary] test> use j_users
switched to db j_users
Atlas atlas-jastie-shard-0 [primary] j_users> db.users.insertMany([
... { "login": "student", "password": "tneduts" },
... { "login": "myuser", "password": "mypas" },
... { "login": "teacher", "password": "qq" },
... { "login": "myking", "password": "myqueen" }
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId("6681e6968d298a95a166c45c"),
    '1': ObjectId("6681e6968d298a95a166c45d"),
    '2': ObjectId("6681e6968d298a95a166c45e"),
    '3': ObjectId("6681e6968d298a95a166c45f")
  }
}
Atlas atlas-jastie-shard-0 [primary] j_users> |
}
Atlas atlas-jastie-shard-0 [primary] j_users> db.users.findOne({ login: "teacher" }).password
qq
Atlas atlas-jastie-shard-0 [primary] j_users> |
```